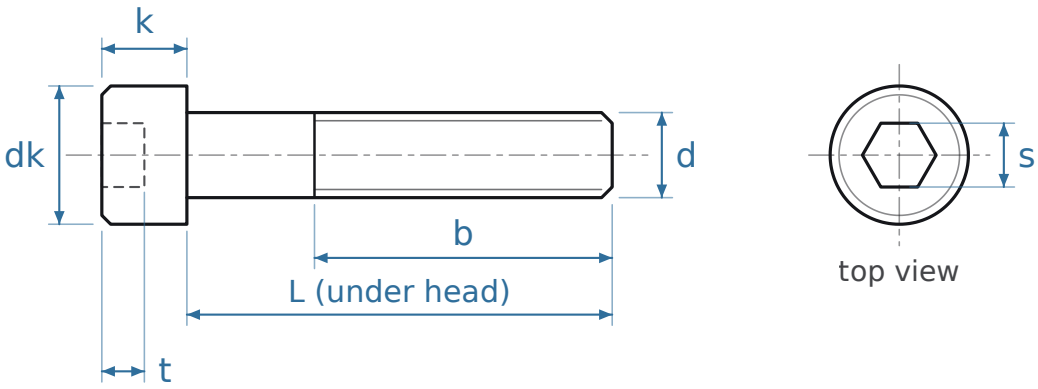
PRINTABLE REFERENCE CHART

Socket head cap screw dimensions: metric ISO 4762, inch ASME B18.3, property classes

BAOUCH.FR
Version of October 16, 2026

1. ISO 4762 socket head cap screw dimensions, M1.6 to M36



Dimensions in millimetres, maximum values for the head (ISO 4762, 2004 edition). P: thread pitch (coarse). dk: head diameter. k: head height. s: hex key size, i.e. the socket width across flats. t: minimum socket depth. b: reference thread length of partially threaded screws.

Screw	P	Head Ø dk	Height k	Key s	Socket depth t min	Thread b ref
M1.6	0.35	**3**	**1.6**	**1.5**	0.7	15
M2	0.4	**3.8**	**2**	**1.5**	1	16
M2.5	0.45	**4.5**	**2.5**	**2**	1.1	17
M3	0.5	**5.5**	**3**	**2.5**	1.3	18
M4	0.7	**7**	**4**	**3**	2	20
M5	0.8	**8.5**	**5**	**4**	2.5	22
M6	1	**10**	**6**	**5**	3	24
M8	1.25	**13**	**8**	**6**	4	28
M10	1.5	**16**	**10**	**8**	5	32
M12	1.75	**18**	**12**	**10**	6	36
M14 (non-preferred)	2	**21**	**14**	**12**	7	40
M16	2	**24**	**16**	**14**	8	44
M20	2.5	**30**	**20**	**17**	10	52
M24	3	**36**	**24**	**19**	12	60
M30	3.5	**45**	**30**	**22**	15.5	72
M36	4	**54**	**36**	**27**	19	84

Head diameters are for a plain head; a knurled head may be slightly larger (10.22 mm instead of 10 mm for M6). The standard goes up to M64. Short screws are threaded up to the head: the limit depends on the size, see the standard or the supplier's catalogue.

2. Inch socket head cap screws (ASME B18.3)

Inch-series socket head cap screws, UNC thread: 1/4-20 reads "1/4 inch diameter, 20 threads per inch". Head: maximum values in inches (mm equivalent in brackets, 1 inch = 25.4 mm). As in metric, the head

height equals the diameter, and the head diameter is 1.5 times the diameter from 1/4 up.

Screw (UNC)	Head Ø A (in)	Height H (in)	Hex key J	Key engagement T min (in)
#4-40	**0.183** (4.65 mm)	**0.112** (2.84 mm)	**3/32"** (2.38 mm)	0.051
#5-40	**0.205** (5.21 mm)	**0.125** (3.18 mm)	**3/32"** (2.38 mm)	0.057
#6-32	**0.226** (5.74 mm)	**0.138** (3.51 mm)	**7/64"** (2.78 mm)	0.064
#8-32	**0.270** (6.86 mm)	**0.164** (4.17 mm)	**9/64"** (3.57 mm)	0.077
#10-24	**0.312** (7.92 mm)	**0.190** (4.83 mm)	**5/32"** (3.97 mm)	0.090
1/4-20	**0.375** (9.53 mm)	**0.250** (6.35 mm)	**3/16"** (4.76 mm)	0.120
5/16-18	**0.469** (11.91 mm)	**0.312** (7.92 mm)	**1/4"** (6.35 mm)	0.151
3/8-16	**0.562** (14.27 mm)	**0.375** (9.53 mm)	**5/16"** (7.94 mm)	0.182
7/16-14	**0.656** (16.66 mm)	**0.438** (11.13 mm)	**3/8"** (9.53 mm)	0.213
1/2-13	**0.750** (19.05 mm)	**0.500** (12.70 mm)	**3/8"** (9.53 mm)	0.245
5/8-11	**0.938** (23.83 mm)	**0.625** (15.88 mm)	**1/2"** (12.70 mm)	0.307
3/4-10	**1.125** (28.58 mm)	**0.750** (19.05 mm)	**5/8"** (15.88 mm)	0.370
7/8-9	**1.312** (33.32 mm)	**0.875** (22.23 mm)	**3/4"** (19.05 mm)	0.432
1-8	**1.500** (38.10 mm)	**1.000** (25.40 mm)	**3/4"** (19.05 mm)	0.495

3. Property classes 8.8, 10.9, 12.9 (ISO 898-1)

Steel socket head cap screws come in property classes 8.8, 10.9 and 12.9. The first number gives the nominal tensile strength (12 → 1,200 MPa), the second the ratio of yield to tensile strength (.9 → 90%). ISO 898-1 values, in MPa:

Class	Diameter	Nominal R_m	Min R_m	Min $R_{p0.2}$
8.8	d ≤ 16 mm	800	**800**	**640**
8.8	d > 16 mm	800	**830**	**660**
10.9	all	1000	**1040**	**940**
12.9	all	1200	**1220**	**1100**

In stainless steel, socket head cap screws are most often A2-70 or A4-70 (ISO 3506-1): A2 or A4 is the austenitic grade, 70 a tensile strength of at least 700 MPa. The tightening torque depends on the class and on friction: see the bolt torque chart.

Reference values taken from the article https://www.baouch.fr/en/blog/vis-chc-iso-4762-dimensions/, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.